VS-09 Infrastructure and Platform Enablement

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VS-09.0 Infrastructure and Platform Enablement

Purpose and Scope

VS-09.0 provides the technical and digital foundation that supports every part of the statistical organization. Where the other streams focus on data, methods, or governance, this one ensures all of those activities can actually **run** — safely, efficiently, and at scale. It delivers the IT environments, data platforms, tools, and integration services that make statistical production possible from collection to dissemination, keeping systems secure, resilient, and compliant while remaining flexible enough to adopt new technologies. The value it creates is **reliable enablement** — turning infrastructure from a background function into an active enabler of the statistical mission.

This document is the **business-layer** view of that enablement (the capability of providing platforms and services). The detailed application and technology architecture is the subject of the SAF's application and technology layers.

A note on numbering: each `VS-0x.y` below is both a value stream stage (*what value* is added) and a business process (*how* it is done); the numbering maps one-to-one. See the key-concepts box above.

Position — a supporting stream that works across the whole chain

VS-09 is not a stage in the production line. It is the **foundation underneath** all of it: every primary and supporting stream runs on the platforms it provides.

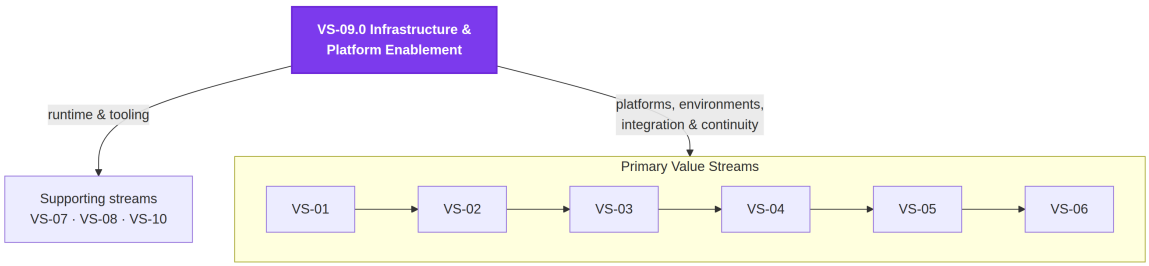


Figure 1: VS-09 is the technical foundation every other stream runs on.

> **!** How a supporting value stream differs from a primary one
>
> - **No steady-state data transition.** VS-09 does not move data along the chain; it provides the environments in which the chain runs.
> - **It runs continuously** as an always-on service, not per production cycle.
> - **Its output is an enabling capability, not a data artefact:** platforms, environments, integration services, and assured availability.
> - **It aligns to GAMSO Capability Management and Corporate Support**, not to GSBPM production phases.

How It Interacts with the Production Chain

VS-09 receives requirements (from business capabilities, data volumes, and the other streams — notably VS-08 for new technology and VS-06 for dissemination platforms) and provides the platforms, integration services, and operational assurance the whole organization depends on. Its security and continuity posture is governed by VS-07.

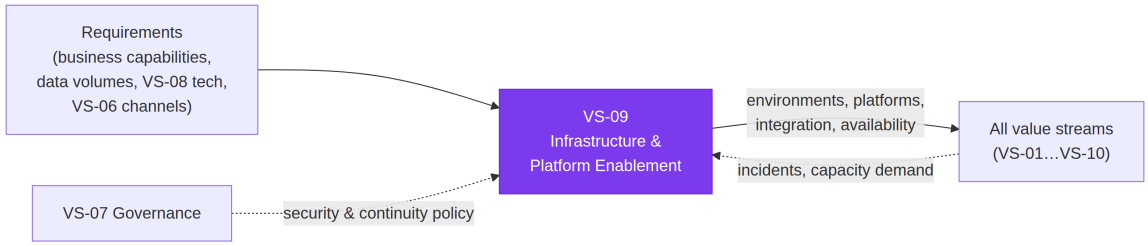


Figure 2: Interaction with the chain: requirements in; platforms, integration, and assured availability out.

Framework Alignment (GAMSO)

Where the primary streams map to **GSBPM** production phases, VS-09 maps to **GAMSO** capability and support activities and the SAF **supporting-business** capability domain (IT):

GAMSO area	Relevance to VS-09
Capability Management	Manage IT capability; introduce and modernize technology
Corporate Support	Manage IT services and operations, security, and business continuity
SAF capability domain: Supporting business (3.x)	Computerization & automation (application development, functional management); security & privacy (technical)

Principles Emphasised

Principle	Statement	How VS-09 applies it
SAF-P10	Security By Design	Builds security and access control into every platform and environment from the start
SAF-P06	Business Continuity	Guarantees availability, resilience, and disaster recovery for the whole chain
SAF-P02	Active support	Runs platforms as actively-supported, monitored services, not one-off builds
SAF-P05	IT and technology is everyone's business	Provides self-service, well-documented platforms so technology serves all teams
SAF-P01	Actively participate in open source to maintain collective control over statistical processes	Favours open, shared platform components to retain collective control

Roles and Actors

Role	Description
Chief Technology / Infrastructure Officer	Accountable for the technical foundation and its strategy
Platform / Cloud Architect	Designs the platform and infrastructure architecture
DevOps / Platform Engineer	Provisions, configures, and operates environments
Integration Engineer	Builds APIs, workflow engines, and orchestration
Security & Continuity Officer	Implements security controls, monitoring, and disaster recovery (with VS-07)
Service Desk / Operations	Runs incident management and keeps services available
Consuming teams (VS-01...VS-10)	Define requirements and run their processes on the platforms

The Value-Stream Flow

VS-09's six business processes form a continuous enablement lifecycle — **define → design → provision → integrate → assure → evolve**.

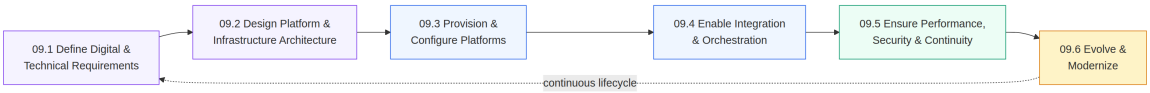


Figure 3: The six business processes as a continuous enablement lifecycle.

Plan — understand what is needed. The lifecycle begins by **defining digital and technical requirements** (09.1): identifying the infrastructure needs arising from business capabilities, data volumes, and analytical complexity across the streams. These feed the **design of the platform and infrastructure architecture** (09.2): blueprints for data storage, processing, integration, and secure access (data fabric, data market, compute platforms).

> **On the Medallion architecture**
>
> A platform team may use Bronze, Silver, and Gold to organize storage, staging, and table optimization inside the platform. That is an engineering pattern, not a governance model: it describes how data has been refined, not what the data means, what methodological state it is in, or what another statistical process may rely on. The steady states remain the governance model; Medallion, if used at all, sits underneath it as a storage convention.

Build — stand it up. The architecture is realized by **provisioning and configuring platforms** (09.3): deploying environments (on-premise, hybrid, or cloud) with the necessary governance, monitoring, and security controls, and by **enabling integration and orchestration services** (09.4): the APIs, workflow engines, and automation that connect the business services and processes — including the steady-state event interfaces the primary streams rely on.

Operate — keep it running and safe. Day to day, VS-09 **ensures performance, security, and continuity** (09.5): monitoring platform performance, implementing data protection, managing incidents, and guaranteeing availability and disaster recovery — the assurance that the whole chain can run reliably.

Improve — modernize. Finally it **evolves and modernizes the technical ecosystem** (09.6): introducing new technologies (often transferred from VS-08), retiring obsolete components, and keeping the foundation flexible for new data sources and operating models.

Business process	What it produces
VS-09.1 Define Digital and Technical Requirements	Infrastructure requirements derived from business needs
VS-09.2 Design Platform and Infrastructure Architecture	Architecture blueprints (storage, processing, integration, access)
VS-09.3 Provision and Configure Platforms	Deployed, controlled environments
VS-09.4 Enable Integration and Orchestration Services	APIs, workflow/automation, event interfaces
VS-09.5 Ensure Performance, Security, and Continuity	Monitoring, security, availability, disaster recovery
VS-09.6 Evolve and Modernize the Technical Ecosystem	Modernization roadmap and adopted technologies

Capability Governance

VS-09 is itself a capability that is **designed** (architecture and operating model), **built** (provisioning and integration), **operated** (continuous monitoring, security, and incident management), and **improved** (09.6). Unlike a primary stream it produces **platforms and assured availability**, not statistical data, and runs as an **always-on service** rather than a per-cycle production run. Its security and continuity are governed by VS-07.

Key Performance Indicators

KPI	Description	Measurement
Platform availability	Measured availability against the agreed service level, per platform	Monthly
Incident response	Mean time to restore service after an incident, per severity class	Per incident, trended quarterly
Event-interface reliability	Delivery rate of the steady-state handover events the chain depends on	Per interface, monthly
Capacity headroom	Share of platforms operating within their planned capacity envelope	Monthly, ahead of known peaks

KPI	Description	Measurement
Security posture	Open critical vulnerabilities and patch latency (assessed with VS-07)	Continuous, reported quarterly
Modernization progress	Components retired or replaced according to the roadmap	Annually

Interactions, Services and Events

Counterpart	VS-09 provides	VS-09 receives	Interaction / event
All streams (VS-01...VS-10)	Environments, platforms, integration, availability	Requirements, capacity demand, incidents	"Platform/service request", "Incident", "Capacity change"
VS-06 Dissemination	Portals, APIs, dashboards, secure research environments	Channel/scaling requirements	"Dissemination platform request"
VS-08 Innovation	Sandbox and production runtime for new tech	New-technology adoption requests	"Technology onboarding"
VS-07 Governance	Security posture and continuity evidence	Security & continuity policy	"Security/continuity review"

Business Objects / Artefacts

VS-09 manages enabling artefacts: **technical requirements**, **architecture blueprints**, **provisioned environments and platforms**, **integration/orchestration services and event interfaces**, **monitoring/security/continuity controls**, and the **modernization roadmap**. These underpin every stream and are managed across their lifecycle (SAF-P03), with security and continuity (SAF-P10/P06) built in. Each artefact has a designated owner; changes are tracked, prior versions remain identifiable, and obsolete components are formally retired.

Applying This to Your NSI

Maturity Assessment Checklist

- ☐ Do we derive infrastructure requirements from business capabilities and data volumes (not ad-hoc)?
- ☐ Do we have architecture blueprints for storage, processing, integration, and secure access?
- ☐ Are environments provisioned with governance, monitoring, and security controls built in?
- ☐ Do we provide integration/orchestration and the event interfaces the streams rely on?
- ☐ Do we monitor performance, manage incidents, and guarantee availability and disaster recovery?
- ☐ Is there a modernization path that adopts new technology (often from VS-08) and retires the obsolete?

Scenarios

Scenario A — Onboarding a secure research environment. VS-06 needs to release de-identified microdata to accredited researchers. VS-09 captures the requirement (09.1), designs the SRE architecture (09.2), provisions it with access controls and output checking (09.3/09.4), and assures its security and availability (09.5) under VS-07's policy — so VS-06 can deliver controlled microdata access reliably.

Scenario B — Adopting a cloud processing platform. VS-08 validates a scalable processing technology. VS-09 designs and provisions the platform (09.2/09.3), enables integration with the existing pipelines (09.4), assures performance and continuity (09.5), and folds it into the modernization roadmap (09.6), so the production streams gain capacity without disruption.

Scenario C — Scaling for the census peak. The census (VS-01) will multiply ingestion and processing volumes for three months. VS-09 derives the peak requirements with the census team (09.1), updates the capacity plan and architecture (09.2), provisions elastic compute and storage ahead of the field period (09.3), and load-tests the ingestion event interfaces end to end (09.4). During the peak it monitors throughput and availability against the agreed service levels (09.5); afterwards the surge capacity is released and the lessons feed the modernization roadmap (09.6). The chain experiences the census as business as usual.